Domain Note

1 Introduction

Metasurfaces are 2-D or planar structures, where the thickness is comparable to the operational wavelength of the light [9]. These artificially engineered structures, depending on the geometry and the material combination, are capable of of exhibiting extraordinary characteristics like perfect absorption, negative refractive index, and so on. They are being utilized sensing, imaging, and tele-communications due to their ability to produce targeted responses like band absorption, beam steering, and polarization [1]. Even though it has immense potential, their design for a specific requirement was a challenge due to the fact the the solution is not unique. This leads to iterative numerical optimization or hit-and-trial methods [2].

Though inverse design addresses these issue through advanced algorithms, inverse design of metasurfaces have been a staggering field due to their complex nature in terms of dependencies between the structure, properties, and the electromagnetic interactions with light. Traditional optimization methods were computational nightmares and were often suboptimal due the issues in the modeling themselves. With the advent of Machine Learning models, the computational time reduced exponentially with deep learning models [3, 6, 4] and generative models entered the domain addressing specific tasks.

Generative Adversarial Networks (GANs) were the first series of model to advance this field through their capabilities in image processing, video analysis, and molecular design. Here, we focus on methods that use images to encode the material and structural parameters for the creating of metasurface designs [8]. The identification of critical issues with GAN like mode collapse and unstable training has cleared way for the newer models to address this issue. One of the most recent works have used a hybrid quantum-classical machine learning approach (QGAN) to reduce the computational time as well as the dataset requirement for the problem [7]. Though this work addresses computational and data requirements, it clearly fails to address the performance advantage of state-of-the-art classical generative models. This is the main motivation behind our study, helping us to aim at Denoising Diffusion Probabilistic Models (DDPMs) that are known for outperforming traditional GANs in the generation of diverse and meaningful outputs[5]. Along with a better architectural design, we also target the improvement of the image encodings and spectral encodings, explained in Section 2.

2 Dataset

The dataset is acquired from a previous study [8] where they have provided the dataset in a public repository[1]. It consists of 18770 images (12,632 metal-insulator-metal and 6,138 hybrid dielectric structures) of size 64×64×3 pixels, representing different metasurface designs.

The images are encoded with planar geometries (G), material properties of the metasurface resonator (M), and the thicknesses of the dielectric layer (T) information. The pixel values in the images are normalized to the range [0, 1], where each pixel's Red, Blue and Green channels correspond to M_1 (plasma frequency for the metal layer), M_2 (refractive index for the dielectric layer), and T (thickness of the dielectric layer), respectively. For MIM structures, the M_2 values are set to 0, while for hybrid dielectric structures, the M_1 values are set to 0. This implies that for MIM structures, the image is Red and Blue combinations, while for hybrid dielectric structures, the image is Red and Green combinations. The thickness information is encoded into the Blue channel of the substrate, though this is semantically inaccurate with respect to pixels. But, we can clearly observe the Geometry information through the contrast Red and Green channels with blue substrate. An example batch of MIM structures are shown in Figure 1.

For each of such images, we have a corresponding spectra (800 points) in the dataset. This spectrum is the absorption spectrum of the metasurface design across a range of wavelengths. The absorption spectrum

[1] `https://github.com/Raman-Lab-UCLA/Multiclass_Metasurface_InverseDesign/tree/main/Training_Data`

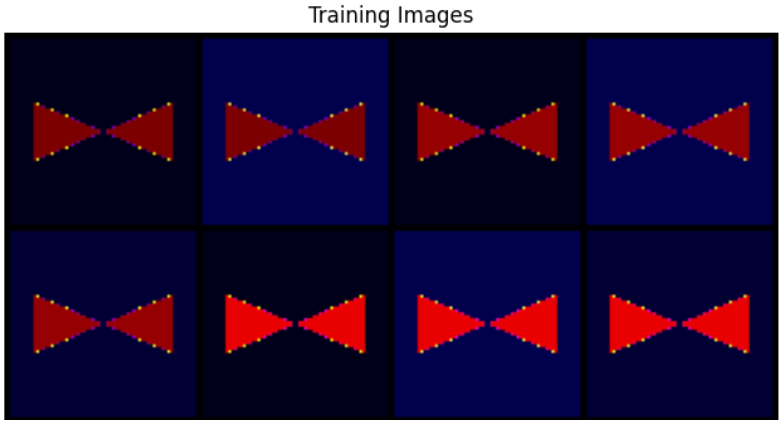

Figure 1: Sample encoded images of MIM structures

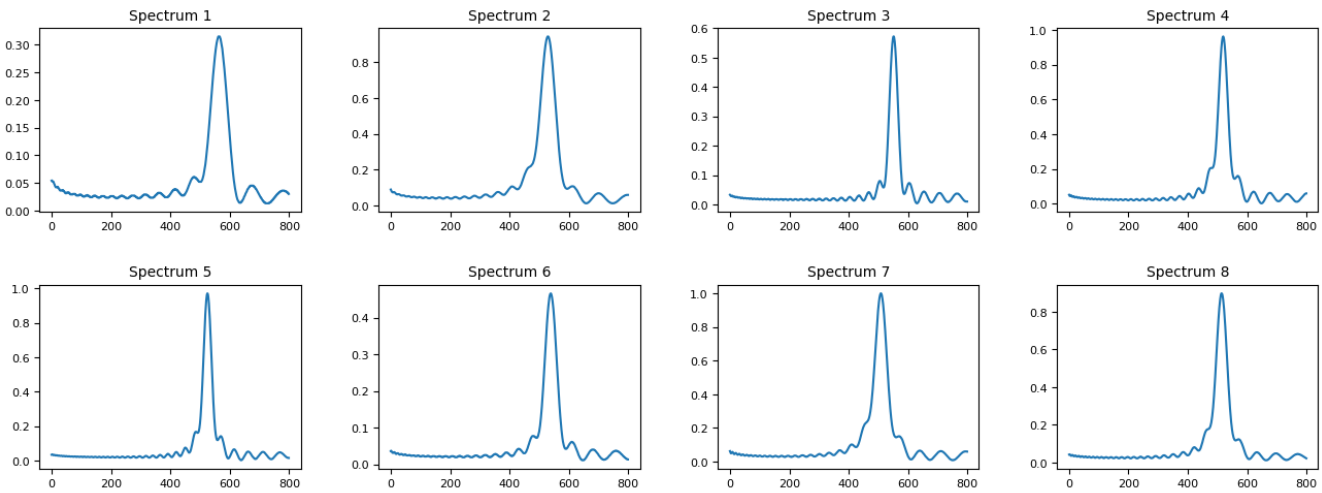

Figure 2: Sample spectra of MIM structures from Figure 1

gives insight in the Q-factor through the amplitude of the absorption peak and the resonant wavelength through the location of the absorption peak. The spectra is normalized to the range [0, 1] as well. An example batch of spectra (corresponding to the above batch of MIM structures) is shown in Figure 2.

3 Problem Statement

Let $\mathbf{s} \in [0,1]^{800}$ denote a normalized absorption spectrum sampled over 800 wavelength points. Let $\mathbf{x} \in [0,1]^{64\times64\times3}$ represent an encoded metasurface design image, where each pixel channel corresponds to material and structural parameters: Red channel $\rightarrow M_1$ (plasma frequency of metal), Green channel $\rightarrow M_2$ (refractive index of dielectric), and Blue channel $\rightarrow T$ (dielectric thickness).

Define a mapping function

$$\mathcal{F} : [0,1]^{800} \rightarrow [0,1]^{64\times64\times3}$$

such that

$$\mathbf{x} = \mathcal{F}(\mathbf{s})$$

The objective is to design $\mathcal{F}$ as the Diffusion Model that generates a physically realizable metasurface design $\mathbf{x}$ whose corresponding absorption spectrum $\hat{\mathbf{s}}$ closely matches the target spectrum $\mathbf{s}$.

References

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